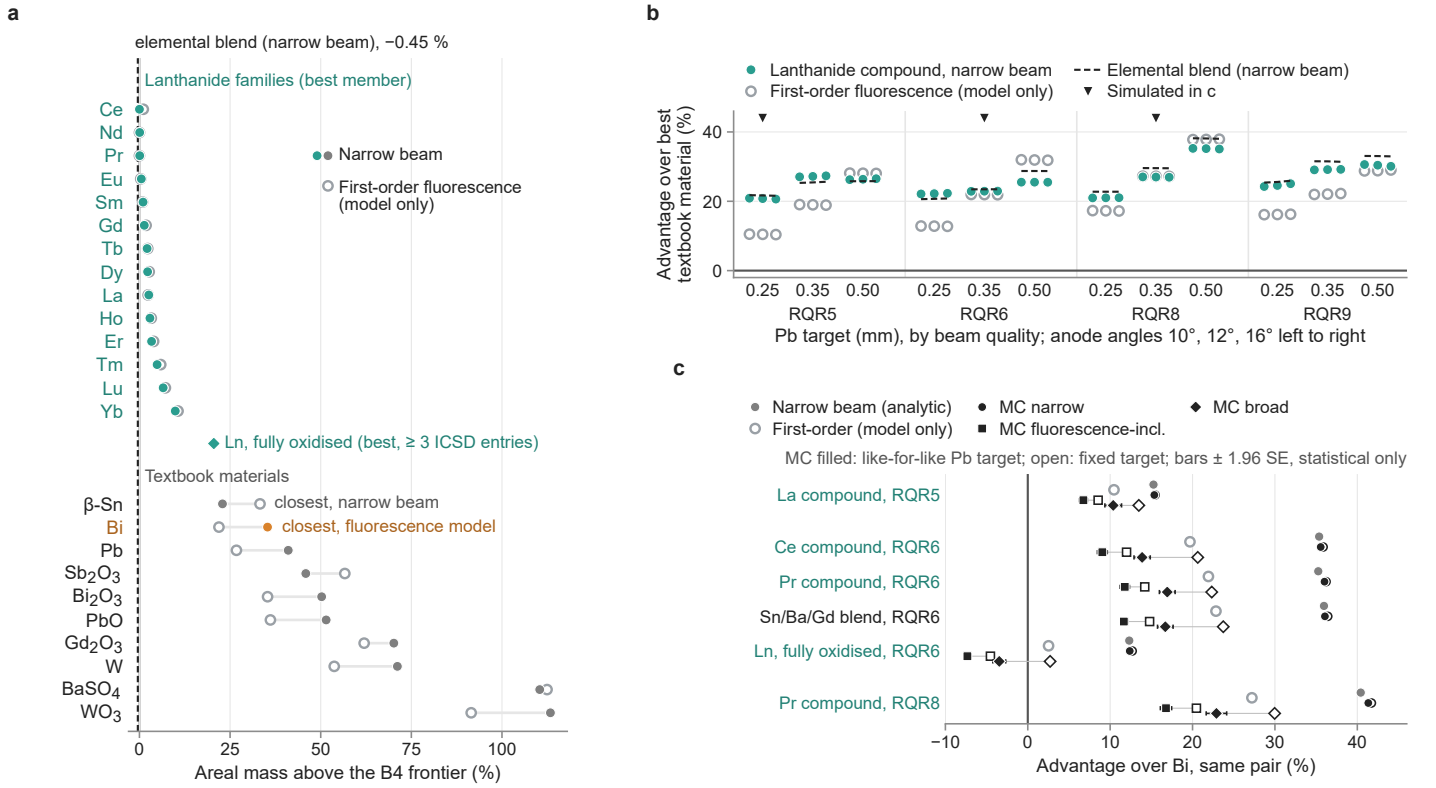

a
elemental blend (narrow beam), −0.45 %
Lanthanide families (best member)
Ce
Nd
Pr
Eu
Sm
Gd
Tb
Dy
La
Ho
Er
Tm
Lu
Yb
Narrow beam
First-order fluorescence (model only)
Ln, fully oxidised (best, ≥ 3 ICSD entries)
Textbook materials
β-Sn
Bi
Pb
Sb2O3
Bi2O3
PbO
Gd2O3
W
BaSO4
WO3
closest, narrow beam
closest, fluorescence model
0
25
50
75
100
Areal mass above the B4 frontier (%)
b
Lanthanide compound, narrow beam
First-order fluorescence (model only)
Elemental blend (narrow beam)
Simulated in c
Advantage over best textbook material (%)
40
20
0
0.25
0.35
0.50
RQR5
RQR6
RQR8
RQR9
Pb target (mm), by beam quality; anode angles 10°, 12°, 16° left to right
c
Narrow beam (analytic)
First-order (model only)
MC narrow
MC fluorescence-incl.
MC broad
MC filled: like-for-like Pb target; open: fixed target; bars ± 1.96 SE, statistical only
La compound, RQR5
Ce compound, RQR6
Pr compound, RQR6
Sn/Ba/Gd blend, RQR6
Ln, fully oxidised, RQR6
Pr compound, RQR8
−10
0
10
20
30
40
Advantage over Bi, same pair (%)